
id: BRIEF-QF-001 uns_version: 0.1 type: brief title: "State of QF Network — Consolidated Assessment Brief (v1.3)" domain: triarch status: ready created: 2026-05-21 author: Ahknu evidence_status: yellow priority: high tags: [brief, qf-network, assessment, consolidation, handoff, decision-support]

pipeline: stage: review next_stage: decision

relationships: consolidates: - "ENG-QF-001" # the engagement - "TEST-QF-001" # the test plan / handoff agenda implements: - "META-CAA-001" # the engagement-type spec references: - "github.com/QuantumFusion-network/qf-solochain @ HEAD 2026-04-21"

x_triarch: purpose: "Single decision-support document for the team. Rolls up all findings to date so others can react. Feeds go/no-go on adopting QF as substrate." decision_owner: Ahknu scope_gate: "ASSESSMENT ONLY — no repair authorized. Adoption decision precedes any code work."

x_fsg: note: "Method (problem-solving rigor) was used throughout. FSG-the-governance-layer is NOT yet applied to QF and would only attach post-adoption, minimally, at the governance/trust layer — never inside the protocol filter logic."

lifecycle: events: - "2026-05-21 created — consolidates ENG-QF-001 + TEST-QF-001 + full repo read" - "2026-05-22 v1.1 — added qf-polkavm-sdk read (embryonic) + one-way claim/bridge finding (engagement-redefining); reordered next-actions, exit-path = remediation priority 1 if adopted" - "2026-05-22 v1.2 — ERC-20 market data pulled (6395 holders, live Uniswap liq); one-way severity narrowed HIGH->MED-HIGH, likely scenario b (no seizure), source confirmation still pending" - "2026-05-22 v1.3 — USER-VERIFIED DexScreener/Go+ data supersedes search: token economics corrected (price \$0.02, liq \$92K, mktcap \$205K, 24h -37%, bleeding); honeypot CLEARED (sells work, not trapped); one-

way severity MED-HIGH->MEDIUM; tokenomics locked (ownership renounced)"

State of QF Network — Consolidated Brief (v1)

Read this first. This is an assessment, not a verdict and not a repair. Every finding carries a confidence flag:  solid /  assumption /  known gap. All code findings are from a static read of the public repo at HEAD (last commit 2026-04-21). Nothing was compiled or run — no Rust toolchain in the assessment environment. The go/no-go decision is the reader's, not the assessor's.

TL;DR (one paragraph)

QF Network is a **technically sound, MIT-licensed, recently-maintained PolkaVM Substrate L1** whose original differentiator (HVM2 parallel compute) was dropped, and whose **engineering team has gone quiet over the last 30–90 days**. The code asset is in better shape than its outside reputation suggests; the *people* risk is the real problem. The central decision is not "is this junk worth saving" — it is **"is this good, maintained-until-recently code whose authors are leaving — and can we get knowledge transfer before they're gone?"** A warm handoff makes this an opportunity; a cold one makes it expensive archaeology.

What changed during assessment (intellectual honesty log)

Three early desk-pass alarms were **overturned by reading the actual repo**. Logged here because the corrections matter as much as the findings:

Early claim (desk pass)	Corrected finding (repo read)
Custom consensus is the scariest risk	 Finality is stock GRANDPA; SPIN is only leader-election
Bus factor ~2	 Historically ~7–8 real engineers (then see team-decay below)
Repo looks thin / dying	 Steady cadence through Apr 2026; competent CI/CD
The <code>TODO(zotho)</code> is a vulnerability-shaped hole	 Not a forgery hole; binding provided by signature check

The bias gate (assessor holds a position in QF) is the reason these were re-checked rather than trusted. It worked.

Findings by area

GREEN — the code asset is solid

- **Block time:** 100ms is real in code (`MILLI_SECS_PER_BLOCK` , "50ms compute / 100ms block"). *Caveat: not yet verified under load — see RED.*
- **Finality:** stock Parity GRANDPA (`pallet_grandpa` wired in qf-runtime). The dangerous surface is NOT homegrown.
- **Architecture:** coherent two-runtime design — `qf-runtime` (fast solo authoring) + `parachain` runtime (Polkadot-anchored shared security).
- **Shared-security bridge:** `pallet-spin-polkadot` is built AND wired (into the parachain runtime), with a working TypeScript finality relayer. Not abandoned.
- **HVM2 residue:** cleanly gone — zero dead code in the execution path.
- **License:** MIT. Free to fork, modify, build commercially. *Code adoption is legally trivial.* (Token / network / trademark / DAO obligations are separate — see RED legal.)
- **CI/CD:** real — 4 workflows incl. containerized release pipeline.
- **Code hygiene:** 35 TODOs across 65 files (low), only 2 panic/unimplemented stubs, named-owner TODOs tracking real upstream-Parity work.
- **Security core (`submit_finality_proof`):** competent GRANDPA verification — relayer auth, `set_id` match, monotonic replay guard, per-precommit signature check, signer dedup, 2/3 supermajority, overflow-safe arithmetic.

YELLOW — assumptions / documented residual questions

- **Test coverage is inverted.** Mature/standard pallets are tested (claim: 31 tests; spin: 6). The **novel security-critical pallets have ZERO tests** (`spin-anchoring` , `spin-polkadot` , `primitives/consensus-spin`). Fixable — additive — see TEST-QF-001 (19 invariants scoped).
- `votes_ancestries` **accepted but never validated** in the finality bridge. Strong hypothesis it is *safe* (the verifier forces all precommits to one identical target via

`MismatchedTargets` , subsuming the ancestry check). Needs a cryptographer/author nod before trusting. One-sentence handoff question.

- **Parity fork is bleeding-edge** (`polkadot-stable2603`). Fresh, but the contract tooling (PolkaVM / pallet-revive) is the least battle-tested substrate, and chasing upstream is an ongoing maintenance tax that only happens with an active team.
- **38 .unwrap() calls** — modest; worth a pass to confirm none are in the runtime execution path.
- **Contract SDK (`qf-polkavm-sdk`) is real but embryonic.** Thin wrapper over Parity's `pallet-revive` : simplified compilation, allocator/panic handler, an `export` macro. Published on crates.io, one working example. Their own roadmap lists UNCHECKED: typed storage API, JS/TS client gen, address prediction, and **deployment/testing tools**. Last commit 2025-12-04 (~5 months stale — quieter than the chain). Verdict: you can deploy native-Rust contracts today, but you'd write against raw `pallet-revive-uapi` for anything real. Workable, NOT a head start. The mature part (pallet-revive) is Parity's, not QF's.

RED — known gaps (NOT guessed; require resources we don't have here)

-  **ONE-WAY VALUE PATH — engagement-redefining, severity narrowed 2026-05-22 (v1.3 user-verified).** The repo contains a `claim` pallet (forked from Polkadot's Ethereum-claims pallet, GPL, 31 tests) that moves value Ethereum→QF. There is **NO outbound path anywhere in the chain repo**. Confirmed one-way . TRAPPED-FUNDS QUESTION NOW RESOLVED  via DexScreener/Go+ security panel (user-supplied, beats prior search data): the ERC-20 is **NOT a honeypot** — Go+ shows Honeypot: No, Can't-sell-all: No, Transfer-pausable: No, Mintable: No, Owner-can-change-balance: No, Ownership renounced: Yes, Owner balance: 0. Live sells are occurring. **ERC-20 holders are NOT trapped — they can sell.** The one-way issue is strictly about value migrated *to QF mainnet* having no return leg ("roach motel"), NOT seizure of ETH-side funds. Exit path remains remediation priority #1 IF adopted (highest rigor, not rushed), but this is no longer a user-harm emergency. Severity: MEDIUM.
- **TOKEN ECONOMICS — confirmed grim (v1.3 user-verified, corrects prior search data).** QF ERC-20 (0x6019...703d) actual figures from DexScreener: price **\$0.02054**, liquidity **~\$92K**, FDV/mkt cap **~\$205K**, ~5,984–6,395 holders, 4 LP holders. 24h: **-37.31%**, sell-weighted (\$17K sell vs \$5.2K buy vol, 36 sellers/25 buyers). (Prior brief versions cited \$1–2.34 / ~\$793K liquidity from search — THAT WAS WRONG; these user-pulled numbers supersede.) Implication: a \$205K-mkt-cap, \$92K-liquidity, actively-bleeding token **cannot fund a multi-dev team** — directly explains the team-decay finding. The "money bs" is real and structural. Note: ownership renounced +

owner balance 0 means no rug lever, but also means **no one can unilaterally fix tokenomics** — it's locked as-is.

- **Team has gone quiet.** Historically ~7–8 engineers; last 90 days collapsed to effectively 1 active (artemiksion), **0 commits in the last 30 days, silent since 2026-04-21**. Senior contributors (Orlov, Alekseev, Pestov, Samat) dark for 90+ days. This is the single most decision-relevant finding and the code cannot resolve it.
- **Does it compile?** Unknown — no Rust toolchain in assessment env. Must run `cargo check` on QF-capable hardware.
- **Does 100ms hold under load?** Unknown — needs a multi-node testnet with real latency and GRANDPA finalizing. Config value ≠ proven property.
- **On-chain reality** — actual validator count, token/treasury balance, is the testnet currently producing blocks. Needs RPC/explorer access.
- **Team reachable for knowledge transfer?** Unknown — human contact. **This is the decider.** (zotho/artemiksion/Khassanov are the names to reach.)
- **Legal / token / DAO transfer** — who owns it, token-holder obligations, clean transfer/fork path. Needs the law/gov partner.
- **PolkaVM contract SDK maturity** — `qf-polkavm-sdk` not yet read; this is the layer a Triarch moat would deploy on. Last big checkable unknown.

The decision this feeds

Go/no-go on adopting QF as Triarch's substrate, framed by one pivotal question:

Can we secure knowledge transfer from the original engineers (esp. artemiksion / Khassanov / zotho) before they are fully gone?

- **If YES (warm handoff):** strong opportunity. Quality MIT code, motivated-to-exit owners, the test gap is scoped and additive, the open questions are sentence-answerable. Adopt, test-harden the two pallets, attach FSG governance minimally at the trust layer.
- **If NO (cold handoff):** expensive archaeology. You inherit a sophisticated custom-pallet / PolkaVM / forked-Parity chain with untested security-critical code and nobody to ask. Proceed only with eyes open and a real Rust/distributed -systems hire budgeted.

Independent of the above: the A6 dependency-decouple recommendation stands — do not anchor Triarch's attestation layer to QF as sole backend. Put it behind an interface; QF

becomes one swappable option, not the foundation.

Recommended next actions (proposed — decision owner picks)

1. **Bridge scenario RESOLVED** (Go+ panel: not a honeypot, sells work, ownership renounced — ERC-20 holders not trapped). Remaining: confirm there is no QF-mainnet→ETH return leg and scope an exit path IF adopted. No longer an emergency. it has no lock/escrow fn and 0x6019...703d holds no large locked balance (market data already points to scenario b / no seizure — confirm at source). Cheap, decisive.
2. **Reach the devs now.** Warm, specific, peer-level. The handoff agenda in TEST-QF-001 (the OQ list) is the script — lead with the GRANDPA/ancestry questions to signal you read the code. Time-sensitive (team going dark).
3. **cargo check on real hardware.** Cheapest hard fact about workability.
4. **On-chain pull** — validators, treasury, live block production.
5. **Legal pass** with the law/gov partner on token/DAO/transfer AND the one-way value-path liability.
6. **If adopted: a sanctioned exit path is remediation priority #1** — ahead of the moat. First in line, highest rigor, not rushed.
7. **Defer** any FSG-governance mapping and all other code repair until after go/no-go.

Confidence on this brief

🟢 on everything readable from the repo (code quality, license, CI, architecture, test gaps, finality-logic soundness on verifiable branches). 🟡 on the ancestry-safety inference (needs cryptographer/author). 🔴 on compile, load behavior, on-chain state, team reachability, legal — all flagged, none guessed.